



جامعة مصر للمعلوماتية
EGYPT UNIVERSITY
OF INFORMATICS

Egypt University of Informatics
Computer and Information Systems
Data Analysis Course

Outing Preferences and Spending Behavior

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Introduction

Understanding how people prefer to spend their free time outside is important for businesses, event planners, and social researchers. People's outing preferences can reflect their lifestyle, personality, and social behavior.

This study aims to explore different outing preferences among individuals and identify patterns in their choices. By analyzing survey data, we can better understand common trends and what influences people's decisions when choosing how to spend their leisure time.

Research Question

What are the most common outing preferences among individuals?

Hypothesis

We hypothesize that Higher costs reduce enjoyment of outings.

Population of Interest:

The population of interest includes university students and young adults who participate in social and recreational activities.

Sampling Method:

We used convenience sampling, where participants were selected based on accessibility (e.g., family, friends, or online respondents).

This method was chosen because:

- It is quick and practical
- Suitable for small-scale academic research
- Allows easy data collection through online forms

Bias Identification:

Several potential biases were considered:

- Sampling Bias: Since most participants are students, results may not represent all age groups.
- Response Bias: Participants may choose socially desirable answers.
- Question Bias: Questions were kept neutral to avoid influencing responses.

To reduce bias:

- Questions were simple and clear
- No leading or suggestive wording was used
- Survey was anonymous

Survey Questions:

- 1.What's your age?
- 2.What is your gender?
- 3.Places preferred for outings
- 4.What place do you go to most often? (Optional)
- 5.How would you rate the price of the place?
- 6.What is the most important factor when choosing a place to go out?
- 7.How often do you go out?
- 8.When do you prefer to go out?
- 9.How much do you spend on an outing?
- 10.Did the price affect how much you enjoyed the outing?

Online survey link:

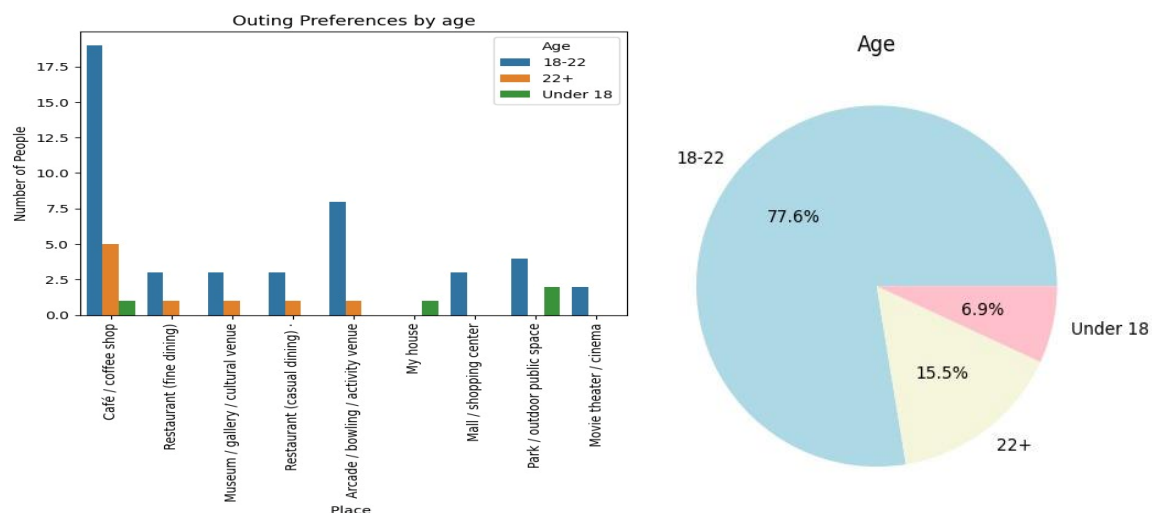
<https://docs.google.com/forms/d/e/1FAIpQLScmpZYbRoX756ZmobIR1vOdeCeOoi7U2-263VNHt3fpmFkUHg/viewform?usp=header>

Number of samples collected: 58

Analysis:

This section presents a detailed analysis of the data collected from 58 participants regarding their social habits and spending behavior. It includes descriptive statistics and visual representations to summarize key patterns in the dataset.

1. Age & Outing Preferences Analysis



Frequency Analysis:

The age distribution shows that most participants belong to the 18–22 age group (77.6%), while the Under 18 (6.9%) and 22+ groups represent a much smaller proportion of the sample. This imbalance explains the higher values observed for the 18–22 group across outing preferences.

Across different outing locations, the 18–22 age group consistently shows the highest engagement, particularly in the most popular venues such as cafés and shopping areas.

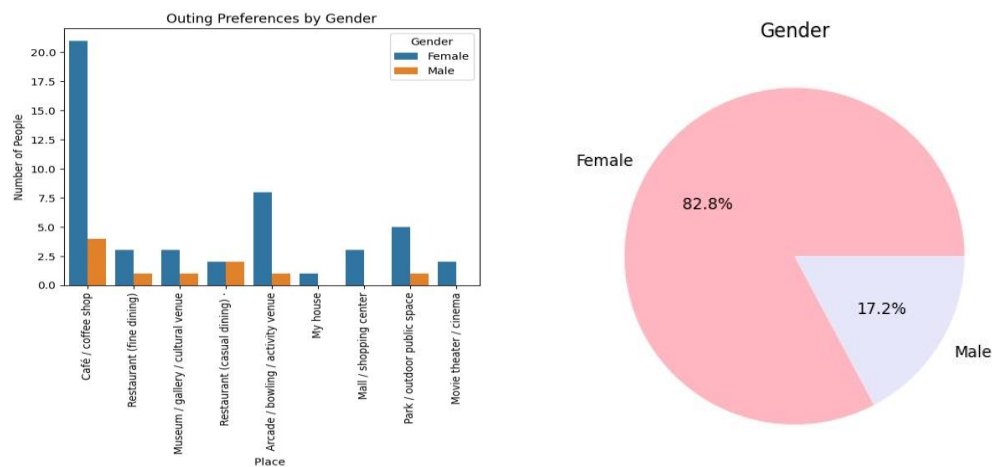
Visualization Description:

Two graphs were used: one showing the age distribution of participants, and another illustrating outing preferences by place across different age groups. Together, they allow comparison between sample composition and behavior patterns.

Interpretation of Results:

The results indicate that the dominance of the 18–22 age group strongly influences overall outing preferences, as this group represents the majority of responses. While smaller age groups show lower values, this is mainly due to their limited representation in the sample rather than lower interest. Overall, outing preferences are primarily driven by the behavior of the 18–22 group.

2. Gender Distribution and Outing Preferences Analysis



Frequency Analysis:

The sample is predominantly female (82.8%), while males represent 17.2%, meaning the results mainly reflect female preferences.

The most popular outing choices are cafés and coffee shops, followed by activity-based venues such as arcades, parks, and shopping malls. These findings indicate a general preference for casual and social environments.

In contrast, fine dining restaurants, museums, and cinemas received lower levels of preference, suggesting reduced interest in more formal or less interactive activities.

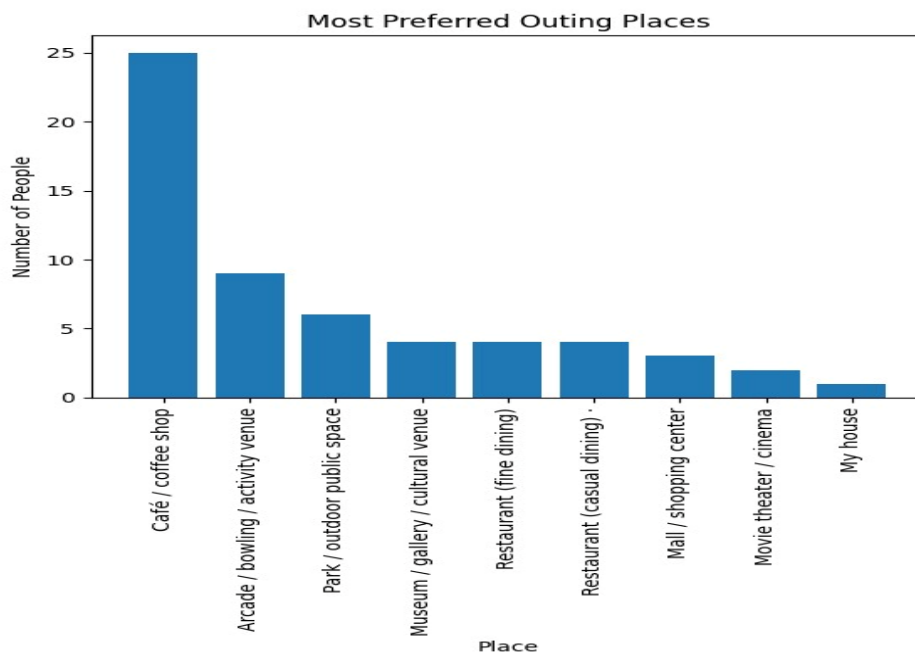
Visualization Description:

A bar chart was used to present participants' preferred outing places. The visualization highlights cafés and coffee shops as the dominant choice across the sample.

Interpretation of Results:

Overall, the results suggest that participants prefer low- to moderate-cost social activities that encourage interaction. Although the sample is mostly female, both genders show similar tendencies toward casual and affordable outing choices rather than expensive or highly structured experiences.

3. Most Preferred Outing Places Analysis



Frequency Analysis:

The results show that cafés and coffee shops are the most preferred outing places, with a significantly higher number of responses compared to other options. This indicates a strong preference for casual and social environments that allow relaxation and interaction.

Activity-based venues such as arcades and bowling centers represent the second most popular choice, followed by parks and outdoor public spaces, showing interest in both entertainment and outdoor experiences.

In contrast, museums, restaurants (casual and fine dining), and shopping malls received moderate levels of preference. The least preferred options were movie theaters and staying at home, indicating a general tendency toward social and interactive outings rather than passive activities.

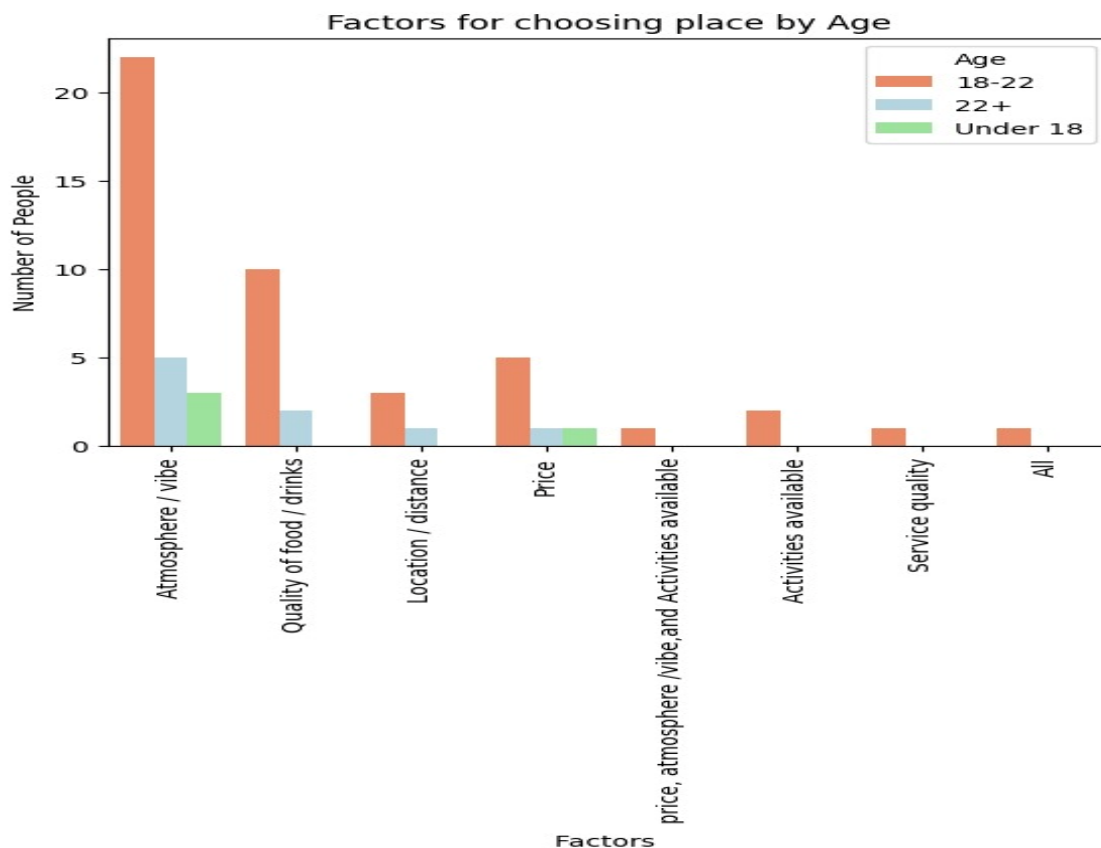
Visualization Description:

A bar chart was used to display the distribution of preferred outing places among participants. The visualization clearly highlights the dominance of cafés and coffee shops over other venue types.

Interpretation of Results:

The results suggest that participants prefer affordable, flexible, and socially engaging locations. In comparison, more formal, expensive, or passive options are less favored.

4. Factors Influencing the Choice of Outing Places



Frequency Analysis:

The criteria for selecting a venue vary, but certain factors are more dominant than others:

- Atmosphere / vibe: The most important factor, selected by 22 respondents from the 18–22 group, 5 from the 22+ group, and 3 from the under 18 group.
- Quality of food / drinks: The second most influential factor, with 10 responses from the 18–22 group and 2 from the 22+ group.
- Price: Selected as the main factor by 5 respondents from the 18–22 group and 1 respondent from each of the other groups.
- Other factors: Including location, activities, and service quality, received minimal responses.

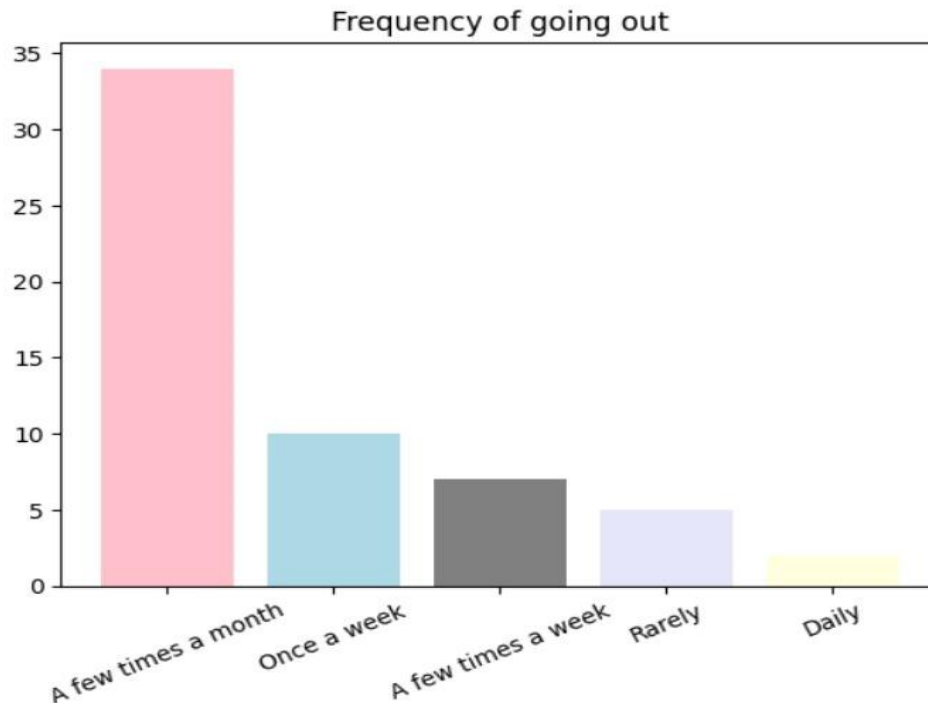
Visualization Description:

A grouped bar chart titled *“Factors for choosing place by Age”* was used to compare preferences across age groups. The visualization clearly shows that atmosphere/vibe is the dominant factor across all groups.

Interpretation of Results:

The results indicate that the overall experience of a venue, particularly its atmosphere and vibe, is more important to participants than practical considerations such as price or location.

5. Outing Frequency Analysis



Frequency Analysis:

The survey results show a clear preference for occasional social activities:

- A few times a month: 34 responses
- Once a week: 10 responses
- A few times a few weeks: 7 responses
- Rarely: 5 responses
- Daily: 2 responses

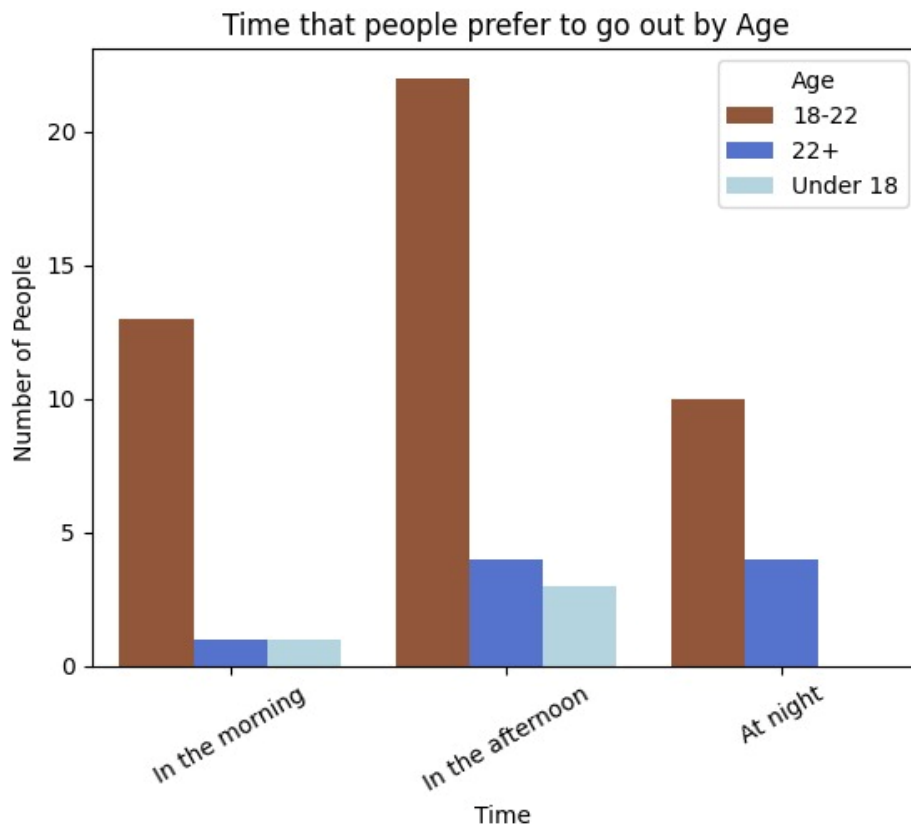
Visualization Description:

A bar chart titled "*Frequency of going out*" was used to display the distribution of responses across different outing frequencies, clearly highlighting the most common behavior patterns.

Interpretation of Results:

The results indicate that most respondents go out a few times a month, suggesting a moderately active social lifestyle. The low number of daily and rare outings shows that most participants fall within a balanced range of social engagement, likely influenced by academic and time constraints.

6. Preferred Time of Going Out by Age Group



Frequency Analysis:

The timing preferences for outings are concentrated in the later part of the day, particularly among the 18–22 age group.

- Afternoon: The most preferred time, with 22 respondents from the 18–22 group, 4 from the 22+ group, and 3 from the under 18 group.
- Morning: 13 respondents from the 18–22 group and 1 respondent from each of the other age groups.
- Night: The least preferred time, with 10 respondents from the 18–22 group and 4 from the 22+ group.

Visualization Description:

A grouped bar chart titled *“Time that people prefer to go out by Age”* was used to compare preferences across age groups. It clearly highlights the dominance of afternoon outings and the relatively low preference for morning times.

Interpretation of Results:

The results show a clear preference for afternoon outings across all age groups, particularly among participants aged 18–22. This may be attributed to greater flexibility during this time of day, making it more suitable for social activities.

7. Spending Patterns on Outings

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round(outing_data["spend_numeric"].mean(),1)
```

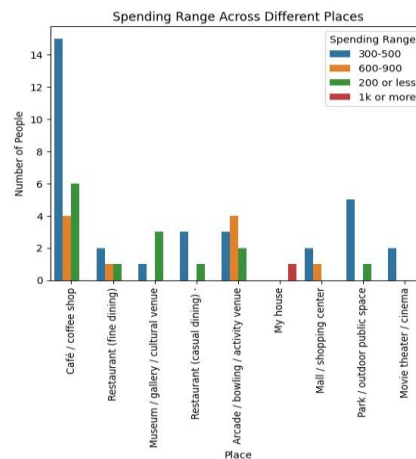
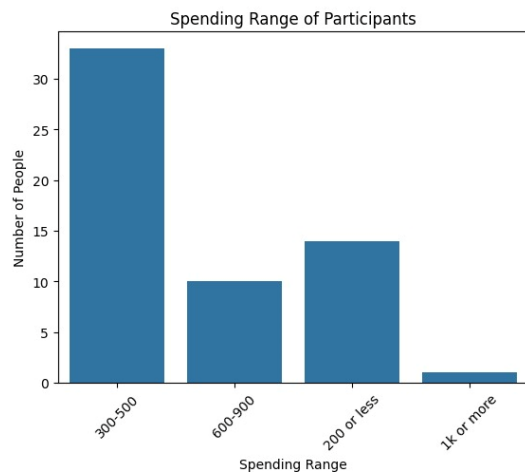
```
np.float64(422.4)
```

```
outing_data["spend_numeric"].median()
```

```
400.0
```

```
round(outing_data["spend_numeric"].std(),1)
```

```
192.7
```



Statistical Summary:

The spending habits of participants were analyzed using measures of central tendency and dispersion:

- Mean: 422.4 EGP
- Median: 400 EGP
- Mode: 300–500 EGP range
- Standard Deviation: 192.7

Visualization Description:

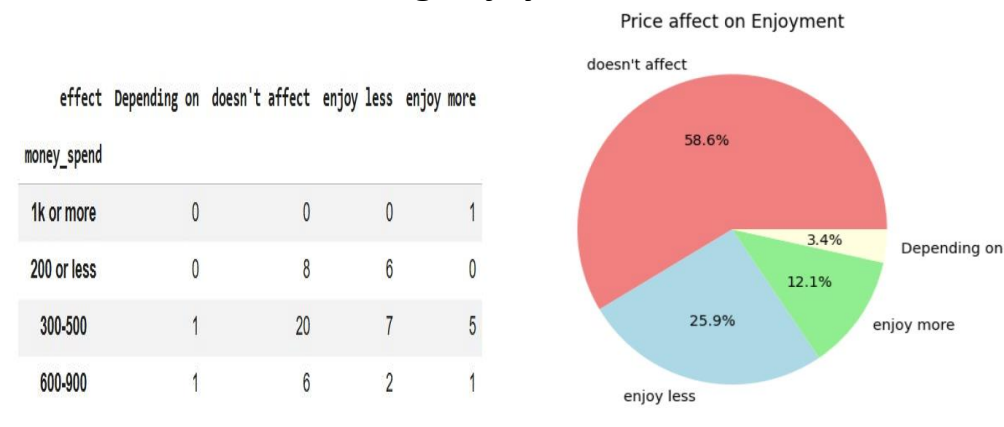
Two visualizations were used:

- A bar chart showing that the 300–500 EGP range is the most frequent spending category.
- A grouped bar chart illustrating spending variation across different venues, with cafés/coffee shops being the most common for the 300–500 EGP range.

Interpretation of Results:

The results indicate an average spending of approximately 422 EGP per outing. The closeness of the mean and median suggests a relatively symmetric distribution, while the standard deviation reflects moderate variation in spending behavior. Overall, most participants prefer mid-range spending, particularly in cafés and leisure venues such as bowling alleys.

8. Effect of Price on Outing Enjoyment



Frequency Analysis:

- Doesn't affect: 58.6%.
- Enjoy less: 25.9%.
- Enjoy more: 12.1%.
- Depending on: 3.4%.

The majority of respondents (58.6%) reported that price does not affect their enjoyment, indicating that the social experience and venue atmosphere are more important than cost. In contrast, 25.9% stated that higher prices reduce their enjoyment, while smaller proportions reported that price increases enjoyment (12.1%) or depends on the situation (3.4%).

A cross-tabulation between spending level and enjoyment showed that participants in the 300–500 EGP range were most likely to report no effect of price on enjoyment (20 respondents). Meanwhile, lower-income respondents (≤ 200 EGP) were more likely to report reduced enjoyment due to cost, while only one respondent in the highest spending category reported increased enjoyment with higher spending.

Overall, the results suggest that once a moderate spending level is reached, price becomes less influential than other factors such as social experience and venue atmosphere.

Conclusion

The results of this study lead to the rejection of the hypothesis that higher costs reduce enjoyment of outings. The majority of participants (58.6%) reported that price did not affect their enjoyment, indicating that cost is not a primary factor influencing overall satisfaction. Although 25.9% of respondents stated that higher costs reduced their enjoyment, this group represents a minority compared to those who were not affected. Additionally, 12.1% of participants reported increased enjoyment when they felt that the outing was worth the price, suggesting that perceived value plays an important role in shaping positive experiences.

These findings highlight that enjoyment is influenced by multiple factors beyond cost, such as the quality of the experience, the environment, and social interactions. Overall, the data suggests that individuals are more focused on the value and overall experience of an outing rather than its financial cost, making cost a secondary rather than a decisive factor.

Any potential issues

This study has some limitations that may affect the results. The sample size was small (58 responses), which limits generalization. Most participants were students, so the findings may not represent all groups. Additionally, convenience sampling and self-reported data may introduce bias and affect accuracy.